Risk Valuation With PUCK Formulation

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Abstract

In financial engineering, ARCH-GARCH models are known as basic tools for risk modeling. We show that these models can be viewed as short correlation limit of PUCK model, which includes the effect of market potential function representing herding behavior of dealers. It is known that PUCK model is very closely related to Random Multiplicative Process (RMP), one of the most important mechanisms of power law distributions. Conventional risk evaluation methods sometimes break down for phenomena following power laws, therefore, PUCK formulation will play an important roles for financial risk evaluation.

Keyword: ARCH-GARCH model, PUCK model, Random Multiplicative Process

1 ARCH-GARCH model

In financial engineering, ARCH-GARCH model is popularly used to formulate market movement $\{X(t)\}$ and its volatility (risk) $\{\sigma(t)\}$ as;

$$\sigma(t+\Delta t)^2 = \sigma_0^2 + c_1\{X(t) - X(t-\Delta t)\}^2 + d_1\sigma(t)^2,$$
$$X(t+\Delta t) = X(t) + \sigma(t+\Delta t)w(t+\Delta t).$$

From this formulation, $\sigma(t)$ depends on its historical movement and the last price difference. In a real market we can observe correlation between price direction, while ARCH-GARCH model does not capture it.

2 PUCK model

Under the formulation of PUCK model, price diffusion in a financial market is described by the following formula with a potential function Φ,

$$X(n+1) - X(n) = -\frac{\partial}{\partial X}\Phi_M(X,n)\Big|_{X=X^*} + G(n),$$
$$\Phi_M(X,n) \equiv \sum_{k=1} b_k(n,M)\frac{X^k}{k}.$$

This discrete process is known to be extended into a continuous formula, equivalent to Langevin equation;

$$\frac{dv(t)}{dt} = -\frac{\mu(t)}{m(t)}v(t) + \tilde{g}^*(t), v(t) \equiv \frac{d}{dt}X(t),$$

$$\frac{\mu(t)}{m(t)} \propto 1 + \frac{b_2(t)}{2}.$$

Second order potential, $b_2(t)$ is estimated from the time sequence.

It is known that PUCK model can be expressed in the following form of random multiplicative process: the price diffusion with discrete model can be described as

$$v(n+1) = b^*(n)v(n) + G(n).$$

The steady state distribution follows a power law, $P(>|v|) \propto |v|^{-\beta}$, in which β satisfies $\langle|b^*(n)|^\beta\rangle = 1$.

We can show that ARCH-GARCH model is special case of PUCK model, which has an independent variable $b_2(t)$ without higher order potential terms.

References

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